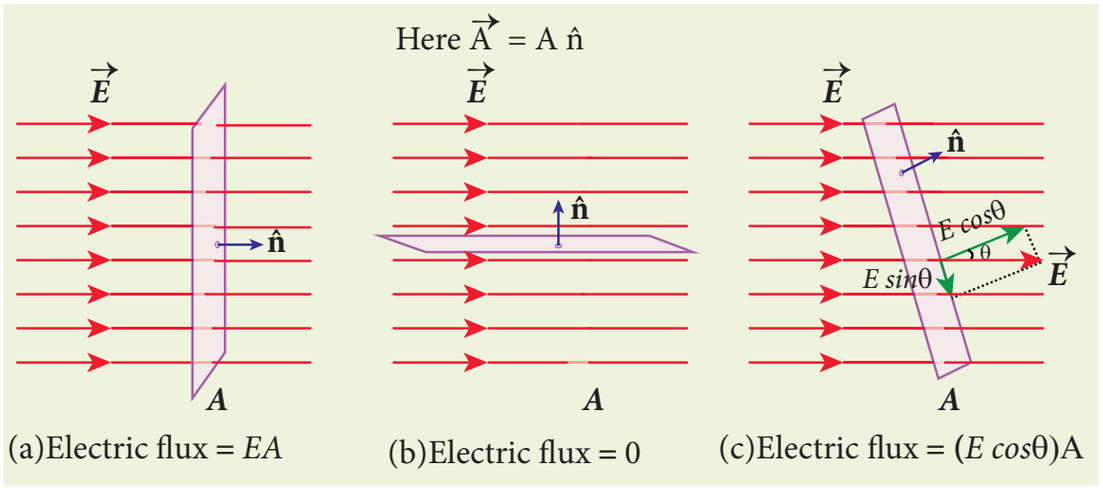
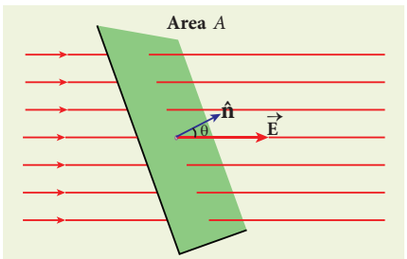


Figure 1.31 The electric flux for Uniform electric field

EXAMPLE 1.17

Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1}. The angle θ is 60°. If θ becomes zero, what is the electric flux?



Solution

The electric flux through the rectangular area

$$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta$$
$$= 100\times5\times10\times10^{-4}\times\cos60^{\circ}$$
$$\Phi_E = 0.25\,\mathrm{Nm^2C^{-1}}$$

For $\theta = 0^{o}$,

$$\Phi_E = \vec{E}\cdot\vec{A} = EA$$
$$= 100\times5\times10\times10^{-4}$$
$$= 0.5\,\mathrm{Nm^2C^{-1}}$$

Electric flux through an arbitrary area kept in a non uniform electric field

Suppose the electric field is not uniform and the area *A* is not flat surface (Figure 1.32). Then the entire area can be divided into *n* small area segments $\Delta\vec{A}_1, \Delta\vec{A}_2, \Delta\vec{A}_3 \ldots\ldots \Delta\vec{A}_n$ such that each area element is almost flat and the electric field over such area element can be considered uniform.

The electric flux for the entire area *A* is approximately written as

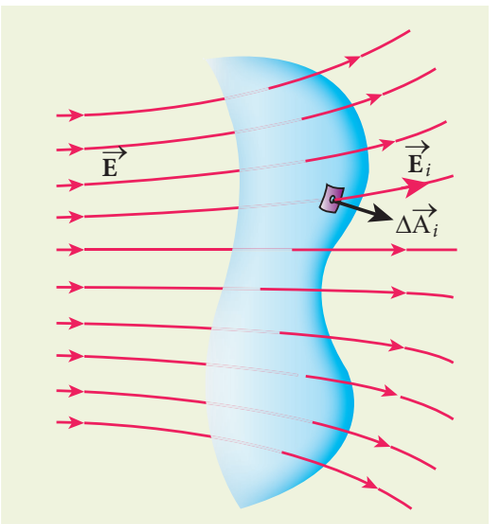


Figure 1.32 Electric flux for non-uniform electric Field

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